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THE MILLISECOND X-RAY FAST SHUTTER FOR BL31 AT ALBA

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Abstract

A new high-speed beam shutter has been developed for the fast x-ray tomography & radioscopy (FAXTOR-BL31) beamline at the ALBA synchrotron, which aims at preventing high dose rate at the sample and provides a synchronization to the acquisition protocol. The non-periodic fast shutter is based on the combination of two tungsten blades each one driven by linear voice coil actuators. The blades synchronization achieves opening and closing times of 10 ms for a monochromatic beam size of H 40 mm x V 12 mm aperture. The design provides flexibility to adjust the aperture dimensions and speed to be able to control the radiation dosage upon the sample, triggered by the image acquisition rate of the detector or timing device. The essential aspects of the design are presented, along with an analysis of the commissioning tests that demonstrate the required performance.

INTRODUCTION

The FAXTOR-BL31 beamline is dedicated to fast X-ray tomography and radioscopy. A portion of its research focuses on the study of biological samples, where it is crucial to ensure precise control over the radiation dose to preserve sample integrity. This necessitates the use of a fast shutter capable of delivering well-defined exposure windows. A key aspect in this particular case lies in the beamline's ability to operate with both white beam and monochromatic configurations, featuring a large beam cross-section of 40 × 15 mm². Blocking such a large beam entirely within a specified time below 10 milliseconds has proven to be particularly demanding, making the selection or development of suitable shutter technology a significant challenge.

TECHNICAL SPECIFICATIONS

Compared to the full beamline capabilities, the technical requirements for the planned research application are narrower in scope and more specifically defined.

The following list represents the most critical constraints considered during the design of the system:

- Monochromatic beam.
- Maximum beam size: 40 × 12 mm².
- Opening/closing time: 10 ms.
- Blade material: Tungsten.
- Minimal blade thickness: 4 mm.
- Beam vertically shutted.

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SYSTEM DESCRIPTION

The solution presented is based on the use of voice coil actuators (VCA) [1, 2]. A voice coil operates through the interaction between a magnetic field and a current-carrying coil, generating a linear force proportional to the input current. This technology enables fast and precise motion, with a high force density that provides the large acceleration required to achieve the desired performance. As shown in the Figs. 1 and 2, the system consists of two VCAs, one positioned at the top and the other at the bottom of the shutter. Mounted on each actuator is a 4.5 mm-thick tungsten blade, which is sufficient to fully block the X-ray beam. It is important to note that the system is symmetric, and each blade is slightly offset relative to the other in the beam axis. This offset allows the blades to overlap in the closed position without the risk of collision, ensuring complete beam blockage.

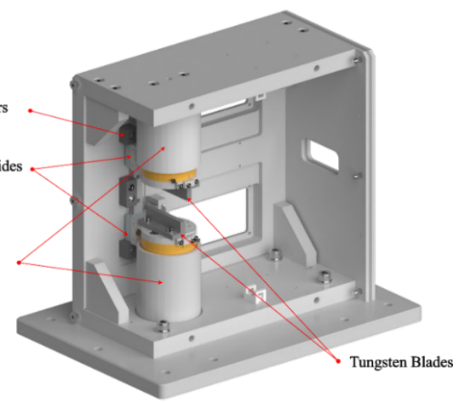

Figure 1: Open view of the shutter.

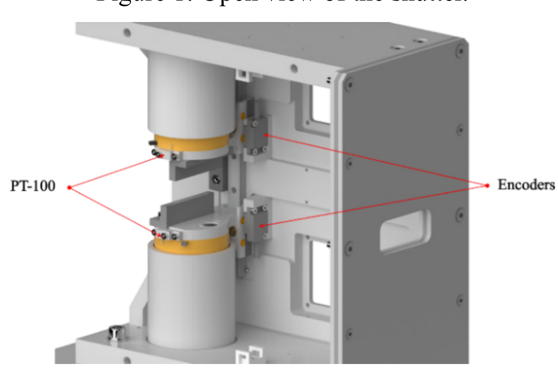

Figure 2: Different sensors provided in the shutter.

The system is equipped with linear guides to ensure accurate linear motion of the blades. In addition, relative encoders are integrated to provide position feedback for each blade. Also, since the blades are mounted directly on the

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same coil of the VCA, and because they will receive radiation from monochromatic beam, PT-100 sensors have been installed to monitor their temperature. Two cooling fans are installed in front of each blade to help dissipate heat if needed.

To protect the surrounding equipment from scattered radiation originating from the slits located upstream of the shutter, a lead plate has been installed as shielding. Additionally, along the beam path, several internal stainless-steel plates have been placed to protect the internal cabling.

Finally, to prevent hard impacts in the event of overshooting the intended motion range, damping elements have been installed just before the actuators reach their travel limits.

Another advantage of using VCA-based positioners is that they allow the aperture size to be adjusted as needed, providing flexibility to the system. In this case, all calculations and design parameters aimed at meeting the 10 ms transition time have been based on an aperture of 12 mm. However, the actuators' range of motion would allow an opening of up to 16 mm, at the expense of a longer opening time. Figure 3 shows the operating scheme of the system, illustrating the beam path as well as the positions of maximum opening and complete closure.

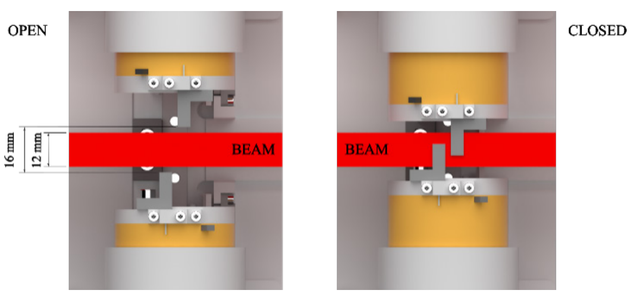

Figure 3: Operating scheme of the shutter.

COMMISSIONING

After the shutter was assembled, a series of tests were performed. The goal was, on one hand, to better understand the behaviour of the VCAs using the manufacturer's software, and on the other hand, to characterize each actuator and fine-tune the control parameters to achieve the desired motion.

Once the blade motions were optimized and properly synchronized, an offline test was conducted to assess whether the opening and closing speeds met the required specifications. For this evaluation, a simple experimental setup was implemented, consisting of a light source on one side and a photosensor connected to an oscilloscope on the opposite side. The blades were set to move from a fully open position of 12 mm to complete closure (0 mm).

Figure 4 presents the test results. The yellow line represents the photons detected by the sensor, while the blue trace corresponds to the trigger signal that initiates the blade movement. As observed, the duration of the transition detected by the photosensor—corresponding to the passage from open to closed state—is approximately within the required 10 ms range.

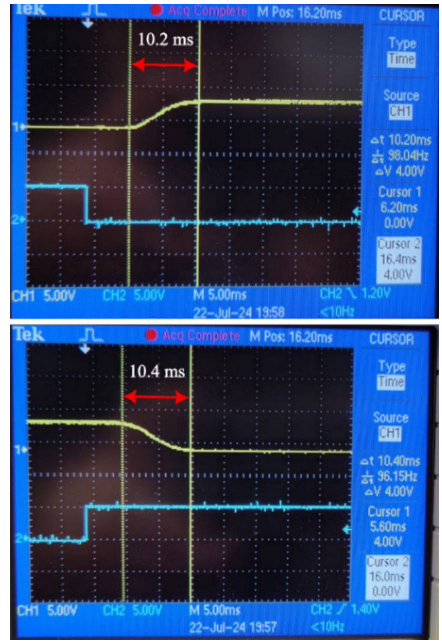

Figure 4: Top graph, opening time. Bottom graph, closing time.

After the system was installed in the beamline and with the aim to obtain a more precise measurement under real operating conditions, the shutter was tested in situ with x-ray beam to verify compliance with the required specifications. For this purpose, the high-speed camera (Phantom S710 camera, Vision Research Inc) was used to capture images during both the opening and closing sequences of the shutter. Figure 5 illustrates these processes. By measuring the number of pixels corresponding to the beam height and counting the number of frames needed to go from fully closed to fully open and vice versa, it was determined that for a displacement of 9.1 mm, the shutter required 6 ms to open and 7.8 ms to close.

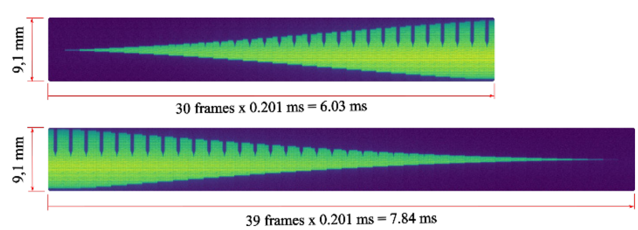

Figure 5: On top, acquired images during opening transition. On bottom, acquired images during closing.

Given the noticeable difference between the opening and closing times, further study and optimization of the system are needed. However, the results clearly demonstrate that the shutter operates within the required specifications and is suitable for its intended application at the beamline.

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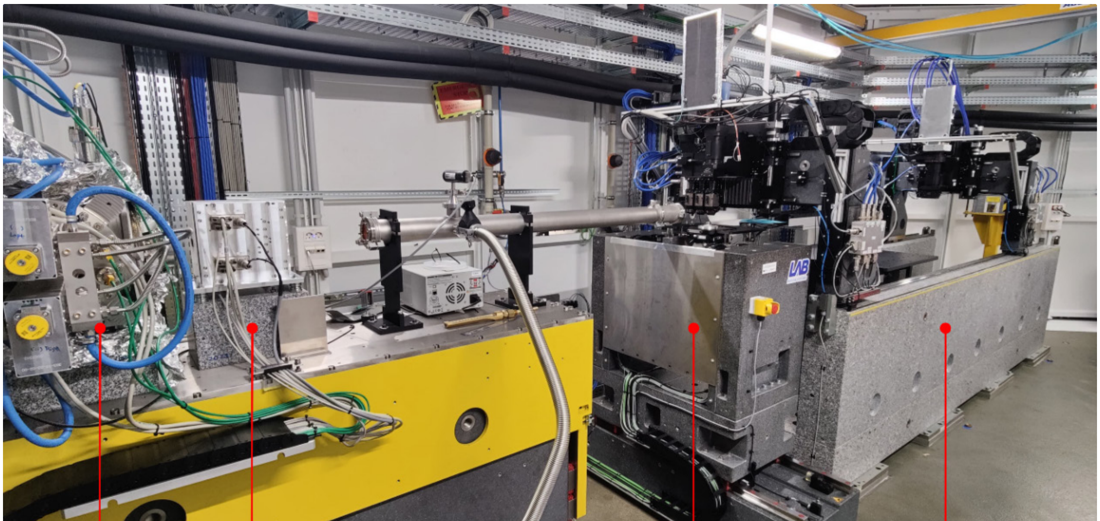

Figure 6: Shutter integrated in the layout of the experimental station of FAXTOR.

BEAMLINE INTEGRATION

The fast shutter is integrated in the experimental area of FAXTOR, positioned just downstream of the final slits and the diamond window that separates the upstream ultra-high vacuum (UHV) section from the ambient air environment in which the shutter operates. Immediately following the shutter are the sample stage and, finally, the detectors. Figure 6 illustrates the integration of the system within the rest of the beamline, while Fig. 7 shows a close-up view of the installed shutter.

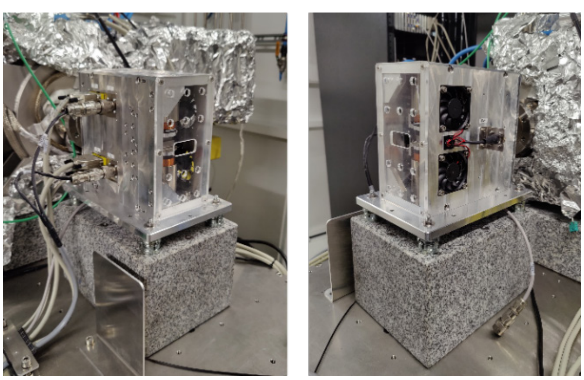

Figure 7: Installed shutter at the beamline.

OPERATION

One operational approach implemented has used the sample stage as the master trigger. In this configuration, the sample stage sends a trigger signal to the shutter to initiate opening. Once the shutter is fully open, it sends a confirmation signal allowing the detector to acquire the images. After a series of images has been captured, the shutter receives a command to close.

Another approach is following a gated acquisition, in which a periodic high-low trigger signal is used to define discrete acquisition windows for the camera.

Considering the reliable performance of the system, it opens the possibility for implementing additional modes of operation. For instance, in scenarios requiring small displacement amplitudes, the shutter could potentially function as a high-frequency chopper, offering fast and repetitive modulation of the beam.

CONCLUSIONS

A fast shutter based on voice coil actuators has been successfully developed and integrated into the FAXTOR-BL31 beamline to address the need for precise and rapid beam control, especially in experiments involving sensitive biological samples. The system achieves beam blocking within the required 10 ms using overlapping tungsten blades, specifically designed to cover a beam section of 40×12 mm². Additional components ensure stable and safe operation.

Tests confirmed reliable performance, and integration into the beamline included appropriate shielding and synchronization with other experimental elements. The shutter supports flexible operation modes, including triggered and gated acquisition, effectively reducing sample exposure while maintaining imaging quality.

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